

Bailey Jannuzzi

PhD Student, Mathematical and Computational Biology
Department of Applied Science
William & Mary

Education

William & Mary Applied Science Department PhD in Mathematical and Computational Biology Advisor: Geoffrey Zahn, PhD	Williamsburg, VA Expected: 2031
Utah Valley University Bachelor of Science in Bioinformatics Magna Cum Laude GPA: 3.81	Orem, UT Spring 2026

Research Experience

William & Mary — TIDAL Lab Graduate Research Assistant Research Advisor: Geoffrey Zahn, PhD Studying microbial community assembly, with a focus on applying graph neural networks to model how microbiomes coalesce. - Contributing to lab research on microbiome-based approaches to plant and ecosystem health, including metagenomics and machine learning methods.	Williamsburg, VA Fall 2026 - Present
Utah Valley University — Hjelman Lab Independent Student Researcher Research Advisor: Carl E. Hjelman, PhD Investigating the diet of forensically relevant blow flies (Diptera: Calliphoridae) to determine their effectiveness as wildlife monitors. - Constructing reproducible bioinformatic pipelines to identify environmental DNA from blow fly gut content samples.	Orem, UT Spring 2024 - 2026
Utah Valley University — Hjelman Lab Peer Research Collaborator Research Advisor: Carl E. Hjelman, PhD Assisting with gathering data on blow fly (Diptera: Calliphoridae) diversity and distribution in Utah. - Conducting field collections. - Preparing collected specimens through pinning and labeling to be identified.	Orem, UT Spring 2025 - 2026

Presentations

*Presenting author

Poster Presentations

American Society for Microbiology, Intermountain Branch Meeting <i>Evaluating Genomic Pipelines for Predicting Bacteriophage Host Range in Salmonella enteric</i> , *Rogers N, Jannuzzi B, Johnston M, Lopez A, Brooks L	2026
--	------

Entomological Society of America, Pacific Branch Meeting 2025
Oxford Nanopore Technology Sequencing and Identification of Fly Gut Content, *Jannuzzi B, Meeds A, Weidner LM, and Hjelman CE

Entomological Society of America, Annual Meeting 2025
From City to Canyon: A Survey of Blow Fly Biodiversity Across Utah Ecoregions, *Beck H, Jannuzzi B, Weidner LM, and Hjelman CE

Oral Presentations

North American Forensic Entomology Association, Annual Meeting | 1st Place Award 2026
From Gut to Database: Leveraging Blow Fly Dietary DNA for Forensic and Conservation Applications, *Jannuzzi B, Hornsby D, Meeds A, Weidner LM, and Hjelman CE

North American Forensic Entomology Association, Annual Meeting 2026
Characterizing Calliphoridae Diversity in the Mountain West: A Statewide Survey of Blow Fly Communities in Utah, *Beck H, Jannuzzi B, Weidner LM, Hjelman CE.

Entomological Society of America, Annual Meeting | 2nd Place Award 2025
Optimizing a Bioinformatic Pipeline for Profiling Blow Fly Diets, *Jannuzzi B, Meeds A, Weidner LM, and Hjelman CE

Entomological Society of America, Annual Meeting 2025
What flies eat and excrete: Insights from puke, poop, and gut contents of blow flies for wildlife and forensic applications, *Weidner LM, Jannuzzi B, Yeilee Perez Ortiz, Gabrielle Meza, Meeds A, and Hjelman CE

Awards and Funding

Funding (Totaling \$3,719)

Undergraduate Research, Scholarly, and Creative Activity Dissemination Grant 2025
Amount Awarded: \$1,093

Scholarly Activities Committee Grant 2025
Amount Awarded: \$1198

Undergraduate Research, Scholarly, and Creative Activity Grant 2025
Amount Awarded: \$1428

Awards and Honors

NAFEA Annual Meeting | 1st Place Oral Presentation 2026

Outstanding Student Award for Biology — UVU, College of Science Spring 2026

ESA Annual Meeting | 2nd Place Oral Presentation 2025

Dean's List — Utah Valley University Spring 2023 - 2026

Experience

William & Mary

Graduate Teaching Assistant | Supercomputing (APSC 624/424)

Williamsburg, VA
Fall 2026

- Grading coursework and assignments
- Assisting students with questions on course material and assignments

Utah Valley University

Teaching Assistant | General Biology (BIOL 1010)

Orem, UT
Fall 2025

- Grading assignments and exams
- Hosting weekly study and review sessions
- Assisting with in-class activities
- Teaching vignette-style lectures in class

Utah Valley University

Teaching Assistant | Genetics (BIOL 3500)

Orem, UT
Spring 2025

- Grading coursework and assignments
- Supporting students in applying class concepts to solve practical problems

Membership

W&M Biomath Journal Club/Seminar

Fall 2026 - Present

UVU Evolution and Bioinformatics Club | President

Spring 2025 - 2026

UVU Evolution and Bioinformatics Club | Secretary

Spring 2024 - 2025

American Society for Microbiology, Intermountain Branch

Spring 2026 - Present

North American Forensic Entomology Association

Spring 2026 - Present

Entomological Society of America

Fall 2025 - 2026

Entomological Society of America, Pacific Branch

Spring 2025 - 2026

Skills

Computational Skills

R, Python, SQL, BASH Scripting/Linux, Bioinformatic Pipeline Development (Oxford Nanopore sequencing, eDNA metabarcoding)

Laboratory Skills

DNA Extraction, PCR, gel electrophoresis, DNA purification, insect specimen preparation (pinning & labeling)